FULL-LENGTH PRACTICE PAPER

Mathematics Class 9 / SSC-I | NBF / Federal Board

Second Annual 2026 Preparation | Total Marks: 75

Purpose: Practice of recurring past-paper skills with changed data and application contexts. This is not a prediction, leaked paper, or official FBISE paper.
Pattern: 15 MCQ marks + 36 short-answer marks + 24 long-answer marks, matching the supplied past/model papers. Use only if this is your applicable examination pattern.

Name: _____ **Roll No.:** _____

SECTION A | 15 marks | Time: 20 minutes

Instructions: All 15 questions are compulsory. Choose one correct option for each question. Each carries 1 mark. For this practice paper, record your answers in the boxes below.

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15

1. The value of $\left(\frac{1}{16}\right)^{-3/4}$ is:

(A) 2 (B) 4 (C) 8 (D) 16

2. The correct scientific notation for 0.00072 is:

(A) 7.2×10^{-4} (B) 72×10^{-4} (C) 7.2×10^{4} (D) 0.72×10^{-4}

3. The domain of $R = \{(1,3),(2,5),(4,3)\}$ is:

(A) $\{3,5\}$ (B) $\{1,2,4\}$ (C) $\{1,3,5\}$ (D) $\{2,3,4\}$

4. The HCF of $12x^2y$ and $18xy^2$ is:

(A) $36x^2y^2$ (B) $6x^2y^2$ (C) $6xy$ (D) $3xy$

5. If $-2x \leq 6$, then:

(A) $x \leq -3$ (B) $x \geq 3$ (C) $x \leq 3$ (D) $x \geq -3$

6. If θ is acute and $\sin\theta = \frac{3}{5}$, then $\cos\theta$ equals:

(A) $\frac{4}{5}$ (B) $\frac{3}{4}$ (C) $\frac{5}{3}$ (D) $\frac{5}{4}$

7. If the bearing of B from A is 075°, the bearing of A from B is:

(A) 105° (B) 255° (C) 285° (D) 345°

8. Two similar triangles have areas in the ratio 9 : 25. Their corresponding sides are in the ratio:

(A) 9 : 25 (B) 27 : 125 (C) 3 : 5 (D) 5 : 3

9. $1 + \cot^2\theta$ equals:

(A) $\sin^2\theta$ (B) $\sec^2\theta$ (C) $\tan^2\theta$ (D) $\operatorname{cosec}^2\theta$

10. The three altitudes of a triangle meet at its:

(A) orthocentre (B) incentre (C) centroid (D) circumcentre

11. Each exterior angle of a regular polygon is 24°. Its number of sides is:

(A) 12 (B) 15 (C) 18 (D) 24

12. The locus of points equidistant from two distinct fixed points is their:

(A) joining line (B) angle bisector (C) perpendicular bisector (D) midpoint only

13. The median of $3, 8, 5, 12, 7$ is:

(A) 5 (B) 8 (C) 12 (D) 7

14. An integer is chosen uniformly from 1 to 20. The probability that it is prime is:

(A) $\frac{2}{5}$ (B) $\frac{1}{2}$ (C) $\frac{3}{10}$ (D) $\frac{1}{5}$

15. An event occurs 18 times in 60 trials. Its relative frequency is:

(A) 0.18 (B) 0.30 (C) 0.60 (D) 0.75

Sections B and C

Time allowed: 2 hours 40 minutes Total marks: 60

Instructions: Attempt every part of Q2 and all three questions Q3–Q5. Choose **one alternative** within each OR pair. Show working and label diagrams. Use your answer book for this practice paper. Round non-exact numerical results to two decimal places unless stated otherwise.

SECTION B | Q2 | $9 \times 4 = 36$ marks

Q2(i) **[4 marks]**

Simplify using the laws of indices: $\dfrac{27^{2/3} \times 16^{3/4}}{8^{1/3}}$.

— **OR** —

For $U = \{1, 2, 3, 4, 5, 6, 7, 8\}$, $A = \{1, 2, 4, 6\}$ and $B = \{2, 3, 6, 8\}$, verify $(A \cup B)' = A' \cap B'$ using sets and Venn diagrams. Complements are relative to U.

Q2(ii) **[4 marks]**

A cyclist travels 12 km due east and then 5 km due south. Draw a labelled sketch and find the three-figure bearing of the cyclist's final position from the starting point. Give the bearing to the nearest degree. (1+3)

— **OR** —

Let $A = \{1, 3, 5\}$ and $B = \{2, 4\}$. (a) List $A \times B$. (b) List $R = \{(x, y) : x \in A,\ y \in B,\ x > y\}$. (c) Find the domain and range of R. (1+1+2)

Q2(iii) **[4 marks]**

Two earthquakes have magnitudes 7.9 and 7.2. Using $M = \log_{10}(I/I_0)$, find how many times the intensity of the first is that of the second. Show the logarithmic steps.

— **OR** —

Find the equation of the line through the intersection of $x + y = 5$ and $x - y = 1$, parallel to $2x - y + 4 = 0$.

Q2(iv) **[4 marks]**

Solve $-3x + 6 \leq 2x + 1 \leq 4x + 9$ and represent the solution on a real number line. (3+1)

— **OR** —

Factorize completely over the real numbers: $(x^2 + 3x + 1)(x^2 + 3x + 3) - 3$.

Q2(v) **[4 marks]**

Prove $\dfrac{1}{1 - \sin\theta} + \dfrac{1}{1 + \sin\theta} = 2\sec^2\theta$, wherever the expressions are defined.

— **OR** —

Each exterior angle of a regular polygon is $18°$. Find (a) its number of sides, (b) each interior angle, and (c) its perimeter if every side is 2.5 cm. (1+1+2)

SECTION B | Q2 continued

Q2(vi) **[4 marks]**

Find the gradient of the line through $A(-2, 1)$ and $B(4, 10)$, then find its equation. (2+2)

— OR —

A bag contains 5 red, 3 blue and 2 green balls. One ball is selected at random. Find the probability that it is (a) blue, (b) not green, (c) red or green. (d) Show that the probabilities of the three colours add to 1. (1+1+1+1)

Q2(vii) **[4 marks]**

A water fountain must be equidistant from $P(2, 3)$ and $Q(8, 7)$. State the geometrical locus and derive its equation. (1+3)

— OR —

Find the equation of a line through $(4, 1)$ perpendicular to $y = -2x+5$. Express your answer as $y = mx+c$.

Q2(viii) **[4 marks]**

Two similar triangles have areas in the ratio $16 : 49$. A side of the smaller triangle is 8 cm. Find (a) the corresponding side of the larger triangle and (b) the ratio of their perimeters. (3+1)

— OR —

A coin is tossed 120 times and heads occurs 42 times. Find the relative frequencies of (a) heads and (b) tails. (c) If the coin is fair, what is the theoretical expected number of heads in 120 tosses? (1+1+2)

Q2(ix) **[4 marks]**

Find the HCF and LCM of $x^2 - 9$ and $x^2 + x - 6$ by factorization. (2+2)

— OR —

In a 50-over match, a team scores at an average of 8 runs per over in the first 10 overs, 6 runs per over in the next 25 overs, and 10 runs per over in the last 15 overs. Find the total runs and the average runs per over for the whole innings. (2+2)

Practice reminder: Choose alternatives question by question. Preparing both sides of one OR pair does not replace preparation for another question.

SECTION C | $3 \times 8 = 24$ marks

Q3 **[8 marks]**

Find **all possible pairs** of real values (p, q) for which $x^4 - 8x^3 + px^2 + qx + 9$ is the square of a quadratic polynomial. Write the corresponding square in each case. (4+4)

— OR —

The vertices of $\triangle ABC$ are $A(0,0)$, $B(6,0)$ and $C(2,4)$. (a) Find the slopes of AB, AC and BC. (b) Find all three interior angles. Check that their sum is $180°$. (3+5)

Q4 **[8 marks]**

From the top of a 30 m tower, the angles of depression of two boats on the **same side** of the tower are $60°$ and $30°$. The boats and the foot of the tower lie on the same straight line at water level. Draw a labelled diagram and find the horizontal distance between the boats. (2+6)

— OR —

Convert $3x + 4y = 12$ into: (a) slope-intercept form, (b) two-intercepts form, (c) symmetric form, and (d) normal form. State the point/direction used in symmetric form and the perpendicular distance from the origin in normal form. (2+2+2+2)

Q5 **[8 marks]**

(a) Construct $\triangle ABC$ with $AB = 6$ cm, $BC = 5$ cm and $CA = 4$ cm. (b) Construct the perpendicular bisectors of all three sides. (c) Label their common point O, name this point, and write the construction steps. (3+3+2)

— OR —

The following table gives the daily electricity consumption of 30 households:

Consumption (units)	0–9	10–19	20–29	30–39	40–49
Frequency	4	6	10	8	2

Using continuous class boundaries and class width 10, calculate the estimated (a) mean, (b) median, and (c) mode. Show the necessary table and substitutions. (2+3+3)

END OF QUESTION PAPER

Prepared by Fakhar STEM Sphere from recurring skill families in the supplied 2025–2026 papers.
Practise methods; exact questions and chapter choices are not guaranteed.